

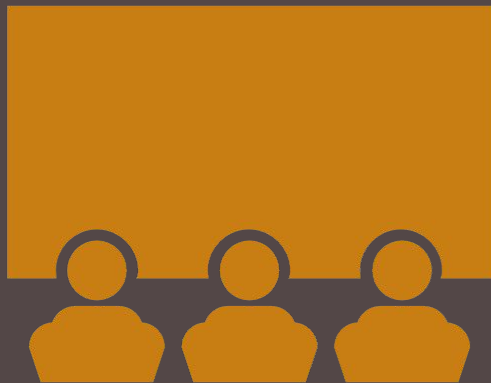
Tech Trends and Developer Workforce Insights

An abstract background graphic featuring a large brown circle in the center. Overlapping this are several smaller circles in shades of gray and brown. In the center of the large brown circle, there are three small yellow dots arranged horizontally, resembling an ellipsis.

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Capstone Project for the IBM Data Analyst Professional Certificate
Justè Gatelytè
June 2026

OUTLINE



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EXECUTIVE SUMMARY

The Goal

I analyzed the 2024 Stack Overflow Developer Survey to provide a data-driven foundation for talent acquisition, workforce training strategies, and technology architecture recommendations.

Key Findings

The industry is moving away from legacy commercial tools in favor of open-source options like Python and PostgreSQL.

Conclusion and Value

The analysis indicates that long-term success requires a dual-focus roadmap: accelerating the adoption of open-source cloud architectures and shifting talent strategies to emphasize employee well-being and inclusive recruitment over salary-based retention.

INTRODUCTION

Report's Purpose

To analyze global developer data to extract actionable insights regarding shifting technology frameworks, database adoption, and changing workforce demographics.

Target Audience

Engineering managers, technical recruiters, and company leadership.

Value

This project replaces guesswork with concrete statistical patterns. It provides the data-driven foundation needed to advise our clients on where to allocate hiring budgets, what skills to train for, and which software architectures will ensure long-term stability.

METHODOLOGY

- **Data Collection** : Pulled data using APIs (parsing JSON directly into Pandas) and performed web scraping using BeautifulSoup and HTTP requests.
- **Data Wrangling** : Used Pandas to inspect data, drop duplicates, and fill missing values. Normalized highly skewed fields (like compensation and experience) using Min-Max scaling, Z-scores, and log transformations so extreme values wouldn't distort the final analysis.
- **Exploratory Data Analysis (EDA)**: Created histograms, KDE plots, box plots, and heatmaps. Removed extreme outliers using IQR and standard deviation thresholds. Mapped the links between pay, remote work, and satisfaction using Pearson and Spearman correlation tests.
- **Visualization & Reporting**: Transitioned from Jupyter Notebooks into Google Looker Studio. Built an interactive dashboard to clearly present the final demographic and technology trends.



Respondent Profiles (MainBranch)

Data Check

- First, we needed to see exactly who answered the survey.

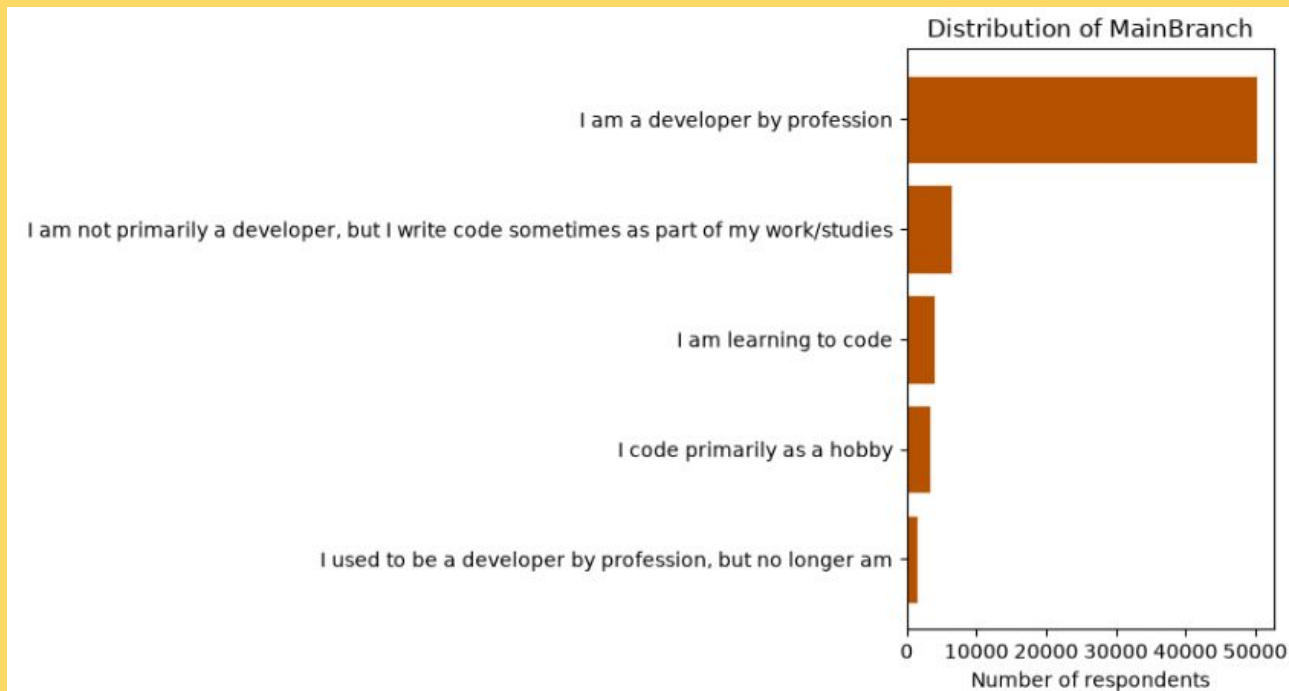
Core Finding

- Most respondents write code professionally for a living.

Why it Matters

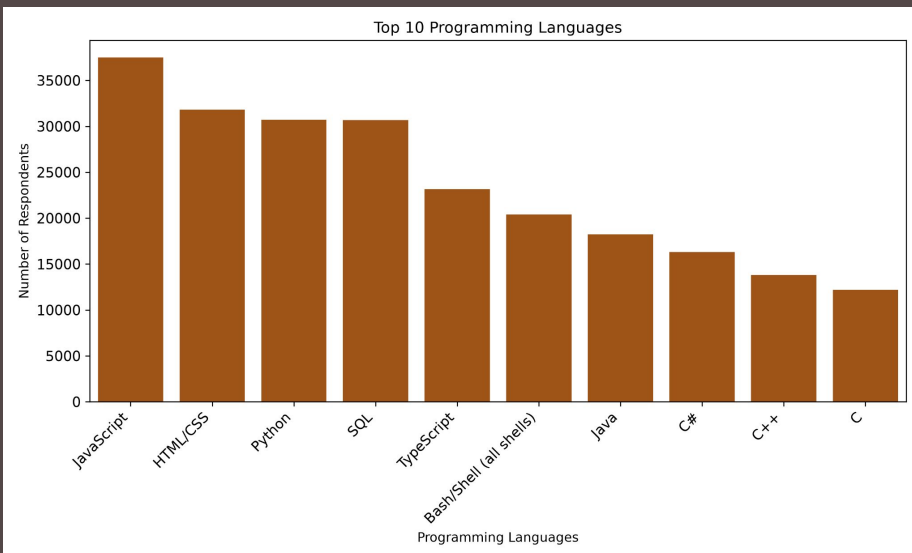
- We are looking at active professionals - our insights can be trusted for corporate strategy.

Respondent Profiles (MainBranch)

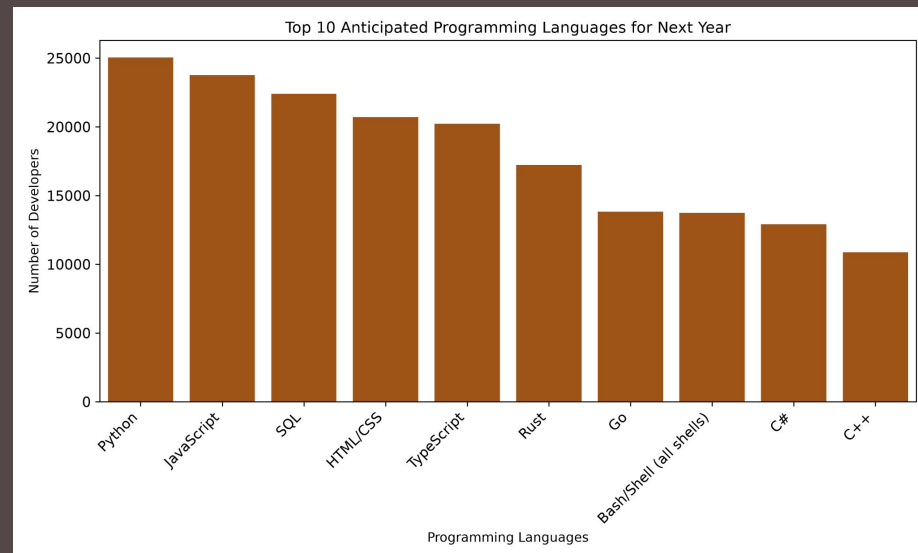


Programming Language Trends

Current year



Next year



Programming Language Trends: Findings and Implications

Current year

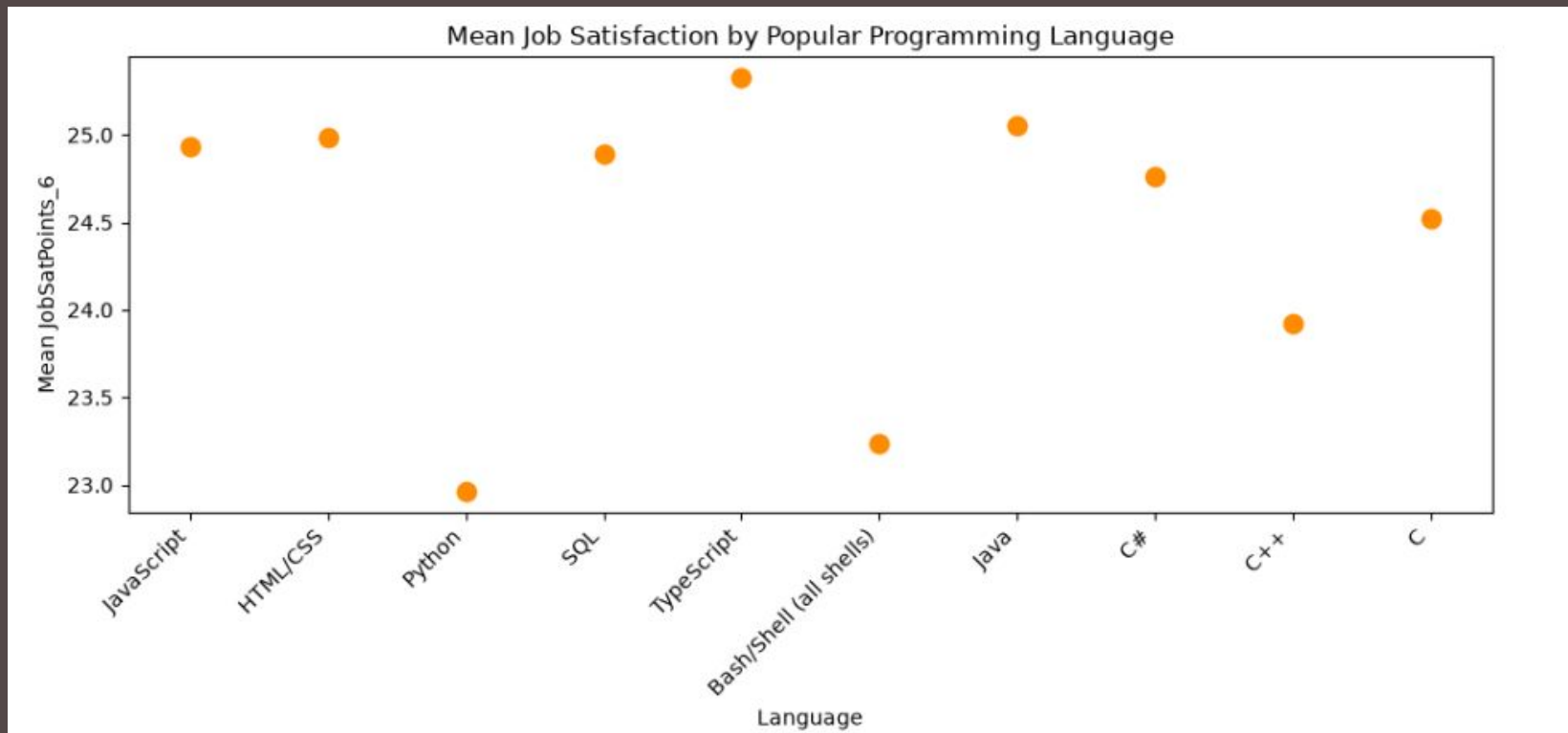
- The Baseline: The data reveals that JavaScript is the undisputed leader in current workflows (exceeding 35,000 respondents), followed closely by HTML/CSS, Python, and SQL.
- The Reality: Traditional web and database tools still run modern businesses.
- The Impact: Companies must continue hiring strong JavaScript and SQL talent simply to maintain their current, everyday operations.

Next year

- The Shift: Python overtakes JavaScript as the #1 language developers want to use next year.
- Emerging Tech: Modern languages like Rust and Go are also seeing massive spikes in interest.
- The Impact: Future technology preferences are shifting toward Data/AI (Python) and fast, secure backends (Rust/Go). Companies should focus their future training budgets here.



The Python Paradox



The Python Paradox

The Expectation

- Python is the #1 language developers across all age groups want to adopt next year.

The Plot Twist

- Paradoxically, current Python developers report the lowest mean job satisfaction.

The Contrast

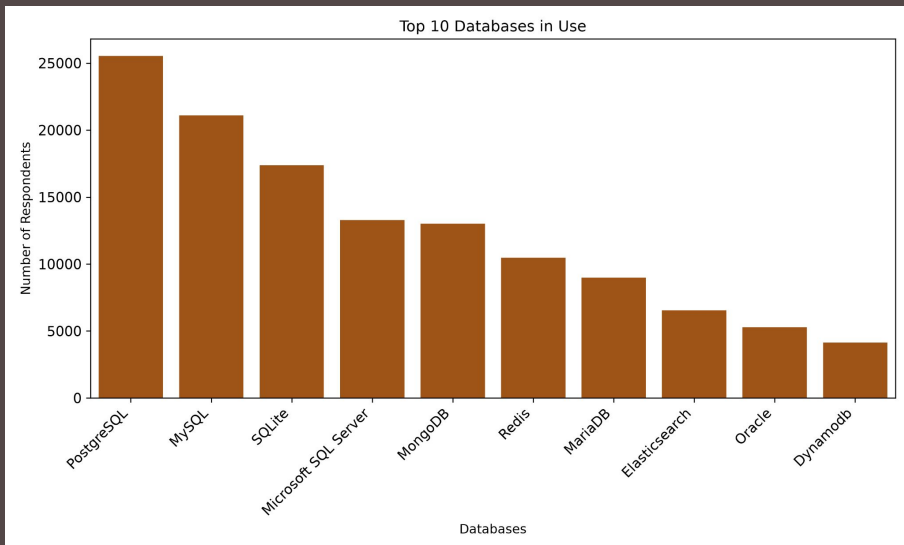
- TypeScript drives the highest satisfaction scores, despite only being the 5th most used and anticipated language.

The Takeaway

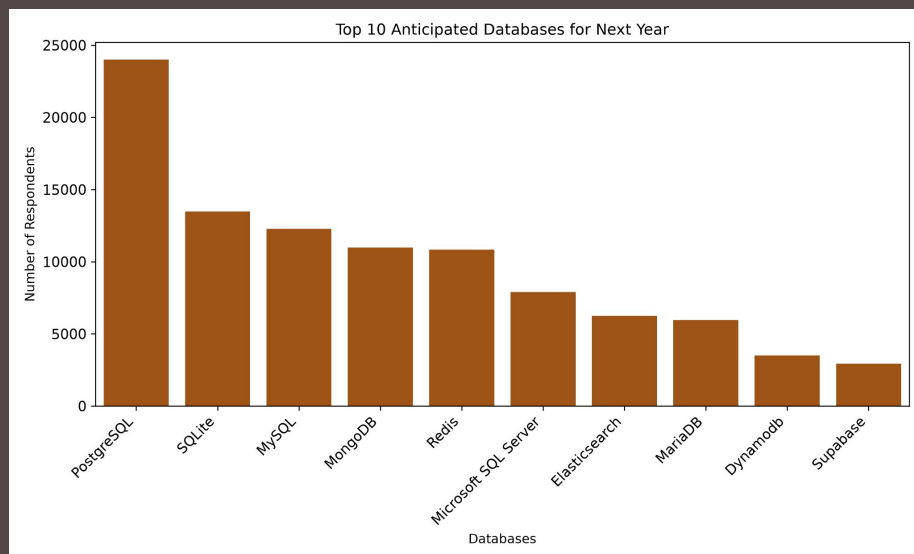
- The hype does not equal daily happiness - trending tech often masks highly frustrating workflows.

Database Trends

Current year



Next year



Database Trends: Findings and Implications

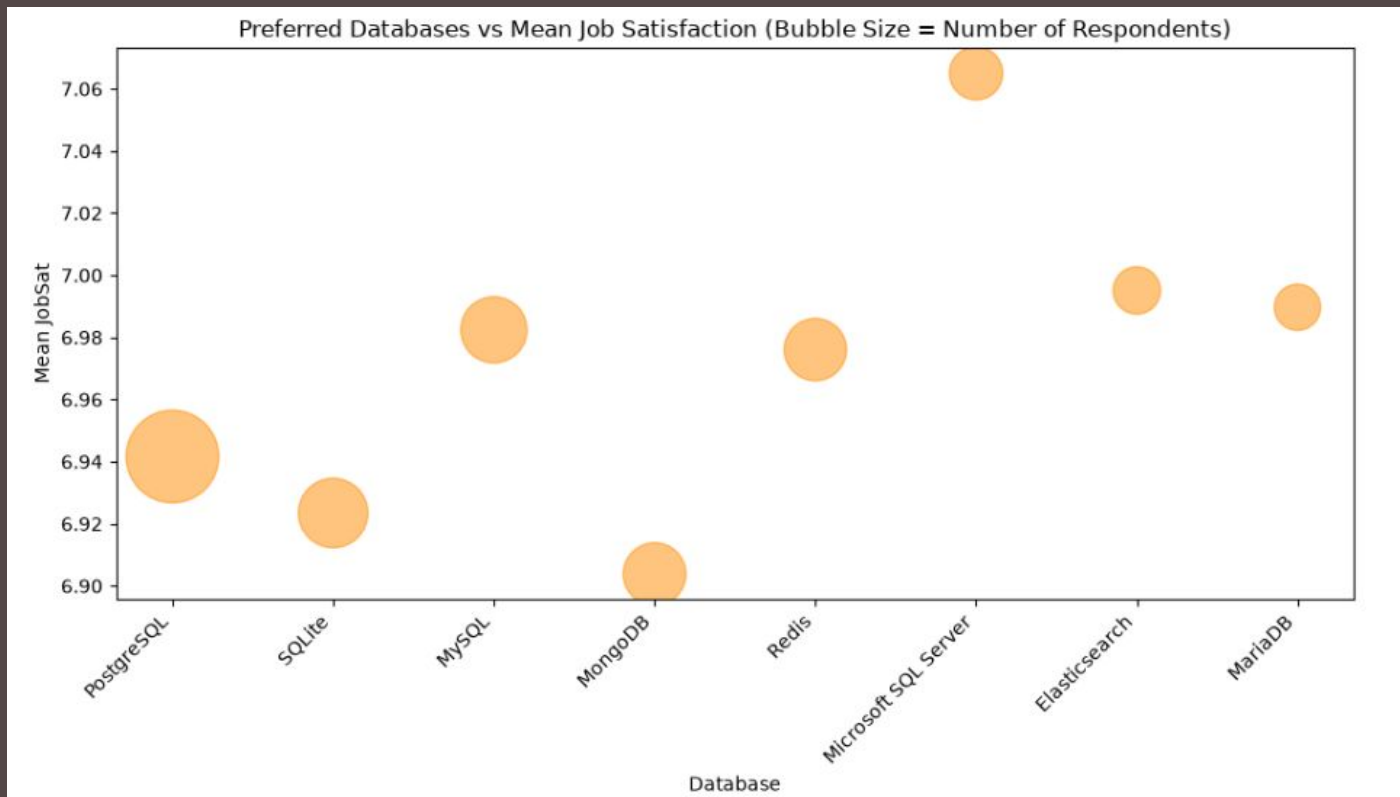
Current year

- The Baseline: PostgreSQL, MySQL, and SQLite are the top three most used databases.
- The Reality: Open-source software now completely dominates over traditional commercial systems (like SQL Server).
- The Impact: SQL skills are still mandatory, but open-source platforms are the clear industry standard.

Next year

- The Shift: PostgreSQL is the #1 database developers want to use next year.
- Emerging Tech: Cloud-native tools (like Supabase) are breaking into the top 10.
- The Impact: The future is open-source and cloud-ready. Consulting clients should turn to PostgreSQL for new projects.

The Database Dilemma



The Database Dilemma

Market Dominance

- PostgreSQL leads in adoption, yet it ranks only mid-tier in overall developer satisfaction.

The Enterprise Advantage

- Microsoft SQL Server achieves the highest satisfaction ratings, despite representing a smaller market segment.

Engineering Friction

- MongoDB reports the lowest satisfaction scores, suggesting potential complexities when managing unstructured data architectures

The Takeaway

- Popularity is not a measure of developer experience; stable, mature ecosystems often outperform trending alternatives in daily satisfaction.

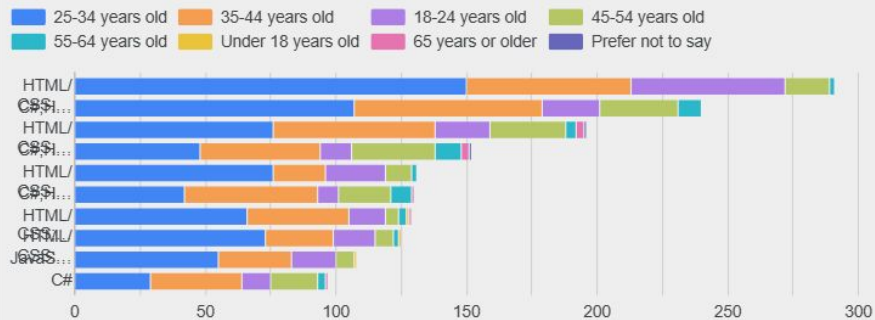
Dashboards



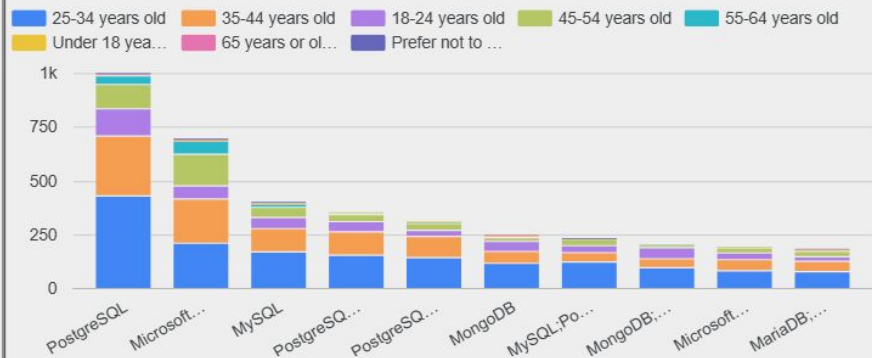
Current technology usage

Country

Top 10 languages used



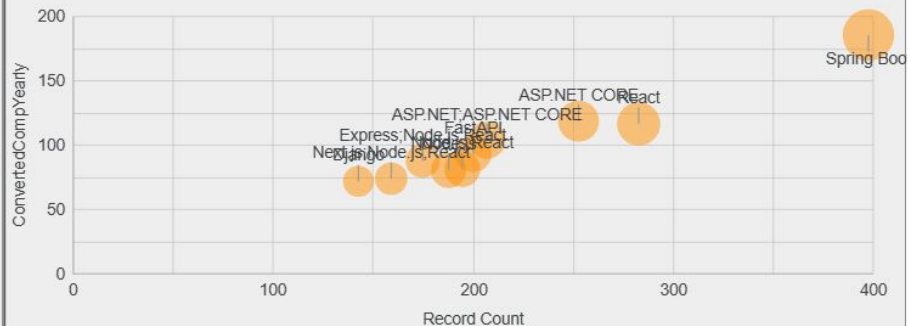
Top 10 databases used



Top 10 platforms used

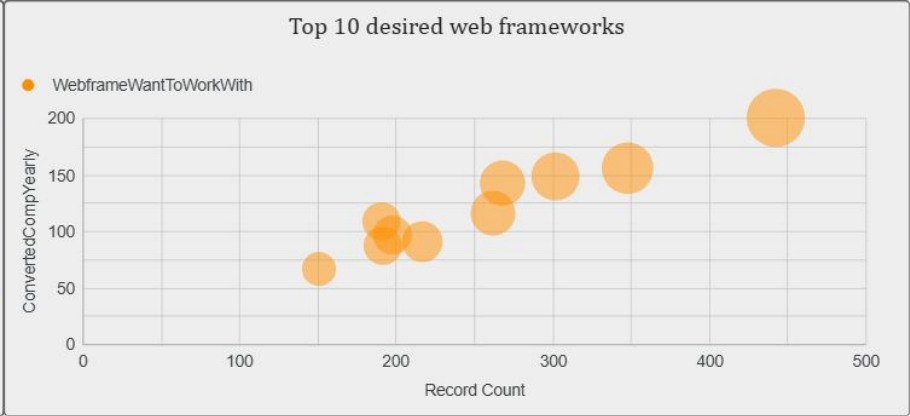
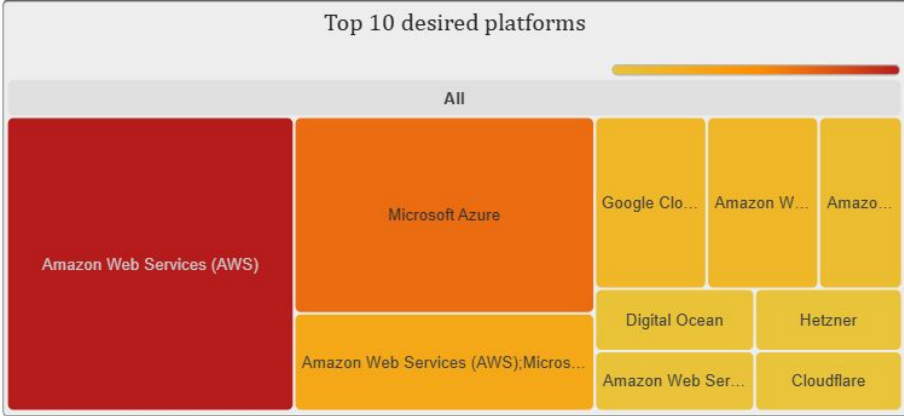
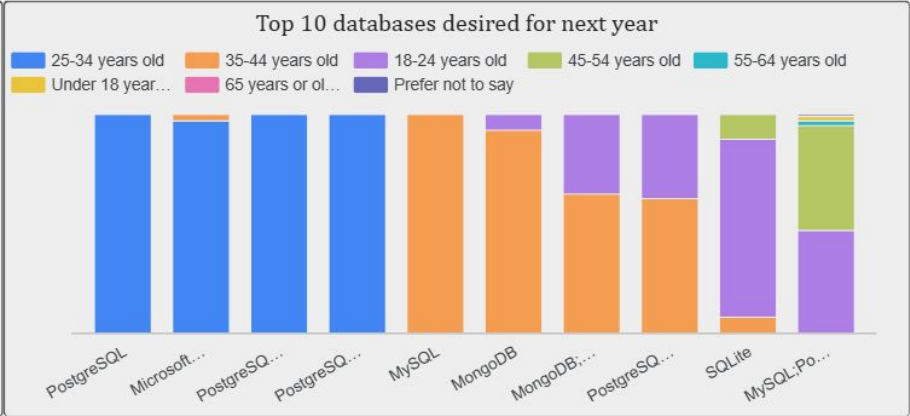
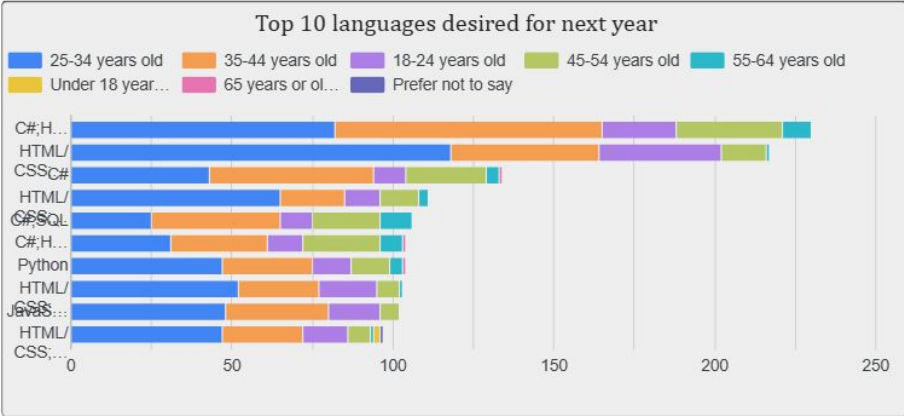


Top 10 web frameworks used



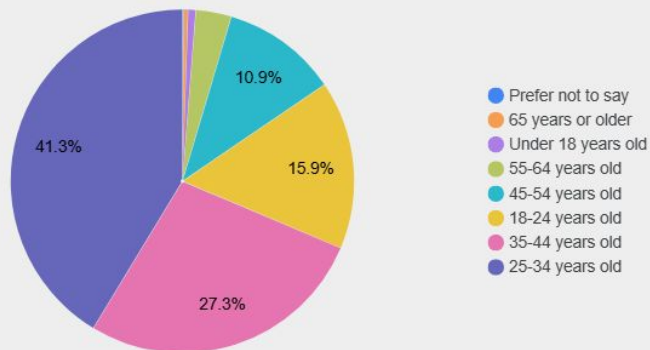
Future technology trends

Country

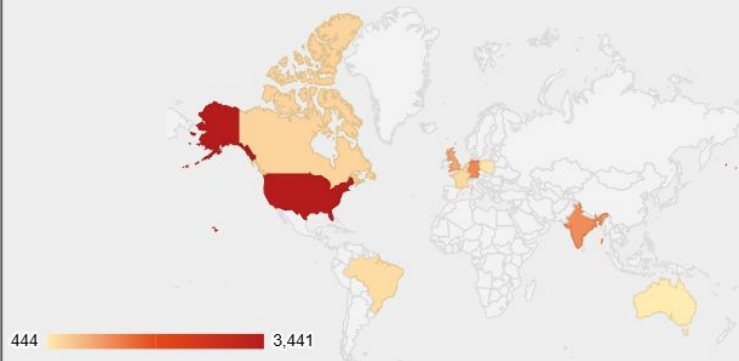


Demographics

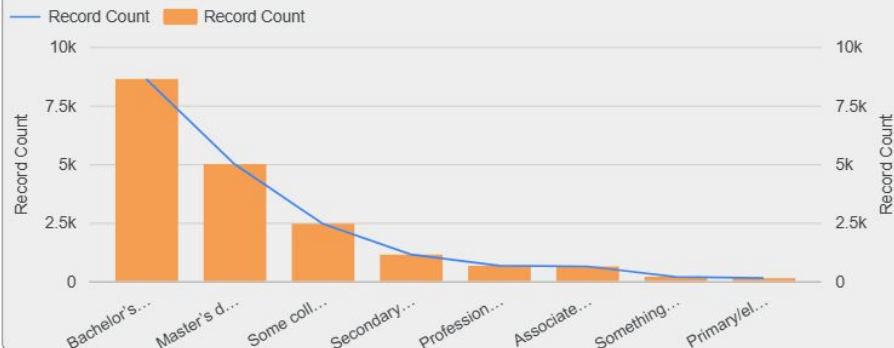
Respondents by age



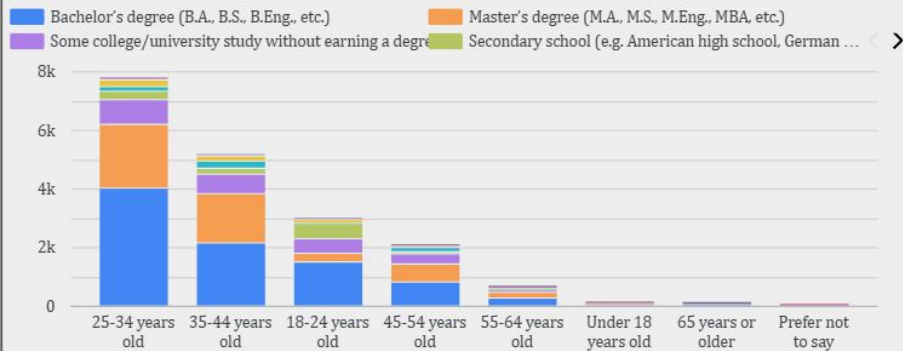
Respondent count by country



Respondent distribution by education level



Respondent count by age, classified by education level



Dashboard Insights

Current Technology Use

Core Tools: Most developers currently use JavaScript and HTML/CSS for building websites.

Age Dynamics: These tools show high adoption across all age groups; notably, the 25-34 age bracket (blue bars) represents a massive portion of the current user base across both languages and databases.

Cloud & Web: AWS and Microsoft Azure are the most used platforms, and Spring Boot is the most popular framework.

Future Technology Trends

The Python Pivot: Developers of all ages want to start using Python next year, likely to build AI or data projects.

Database Choice: PostgreSQL is the top choice for the future, as many developers prefer it over older, expensive commercial databases.

New Tech: Younger developers (18–34) are leading the trend toward newer, flexible cloud tools like Supabase.

Demographics

Who they are: The workforce is young, with nearly 70% of all developers aged 25–44.

Where they are: The global talent pool is heavily concentrated in the United States and India.

Their Education: The industry is highly credentialed, with the vast majority holding a Bachelor's or Master's degree.

Overall Findings and Implications

Overall Findings and Implications

Findings

- The Tech Shift: Python is replacing JavaScript as the #1 language developers want to use next year to build AI and data projects.
- The Satisfaction Paradox: High demand doesn't equal happiness. Trending tools (like Python and MongoDB) actually report the lowest daily developer satisfaction.
- Open-Source Wins: Traditional commercial systems are losing ground; PostgreSQL and cloud platforms like AWS are the clear industry standards.
- The Talent Pool: The global workforce is young (70% are aged 25-44), highly credentialed, and heavily concentrated in the U.S. and India.

Implications

- Invest in AI Skills: Shift training budgets to Python, Rust, and Go to meet future client demand.
- Fix the Friction: Invest in Developer Experience (DevEx) so teams actually enjoy using new tools, rather than just forcing adoption.
- Modernize the Stack: Guide clients away from expensive legacy databases and toward open-source cloud systems.
- Rethink Recruiting: Update hiring strategies to attract young, highly educated talent from global tech hubs.

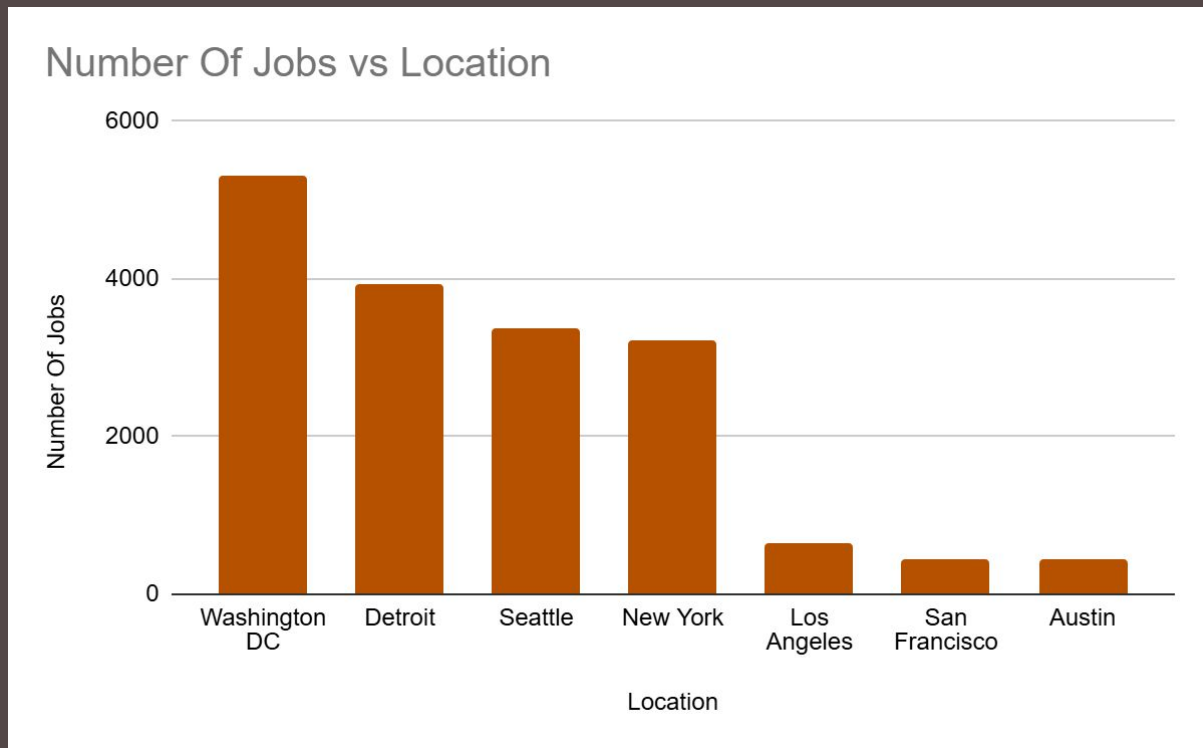
Conclusion

- The Future is Open: Open-source and cloud-based systems are no longer just trends; they are the new industry standard.
- Beyond the Paycheck: Keeping top developers around takes more than just a competitive salary - companies have to actually build a healthy culture and inclusive teams.
- The Bottom Line: To stay competitive, companies need to adapt both the technology and hiring strategies to fit what this young, global workforce actually wants.

Thank you



Appendix: Job Postings



Appendix: Popular Languages



Links

- [GitHub repository](#)
- [Google Looker Studio dashboard](#)